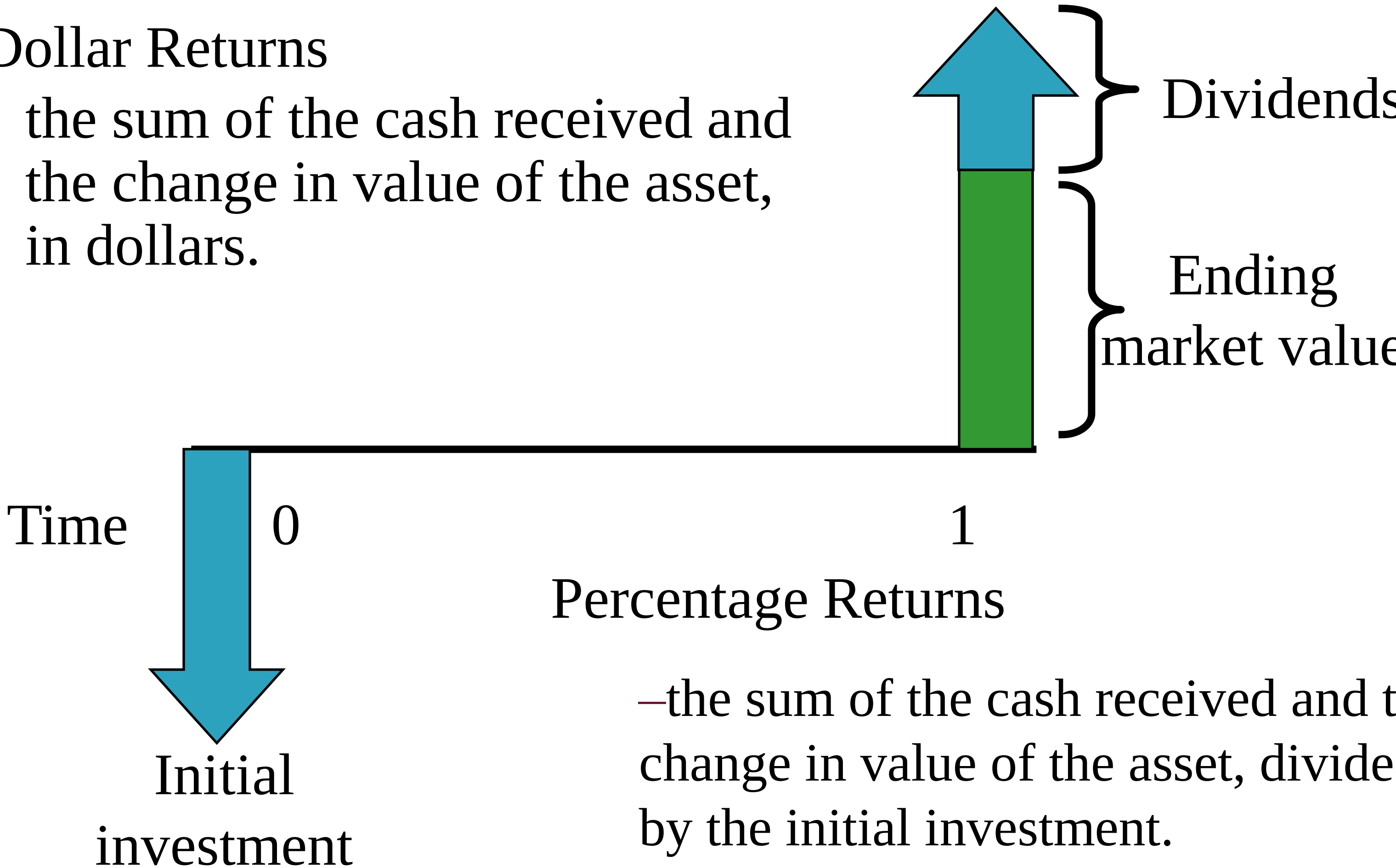
# Risk & Return

# Returns

- Dollar Returns

  the sum of the cash received and the change in value of the asset, in dollars.




Percentage Returns

–the sum of the cash received and the change in value of the asset, divided by the initial investment.

# Returns

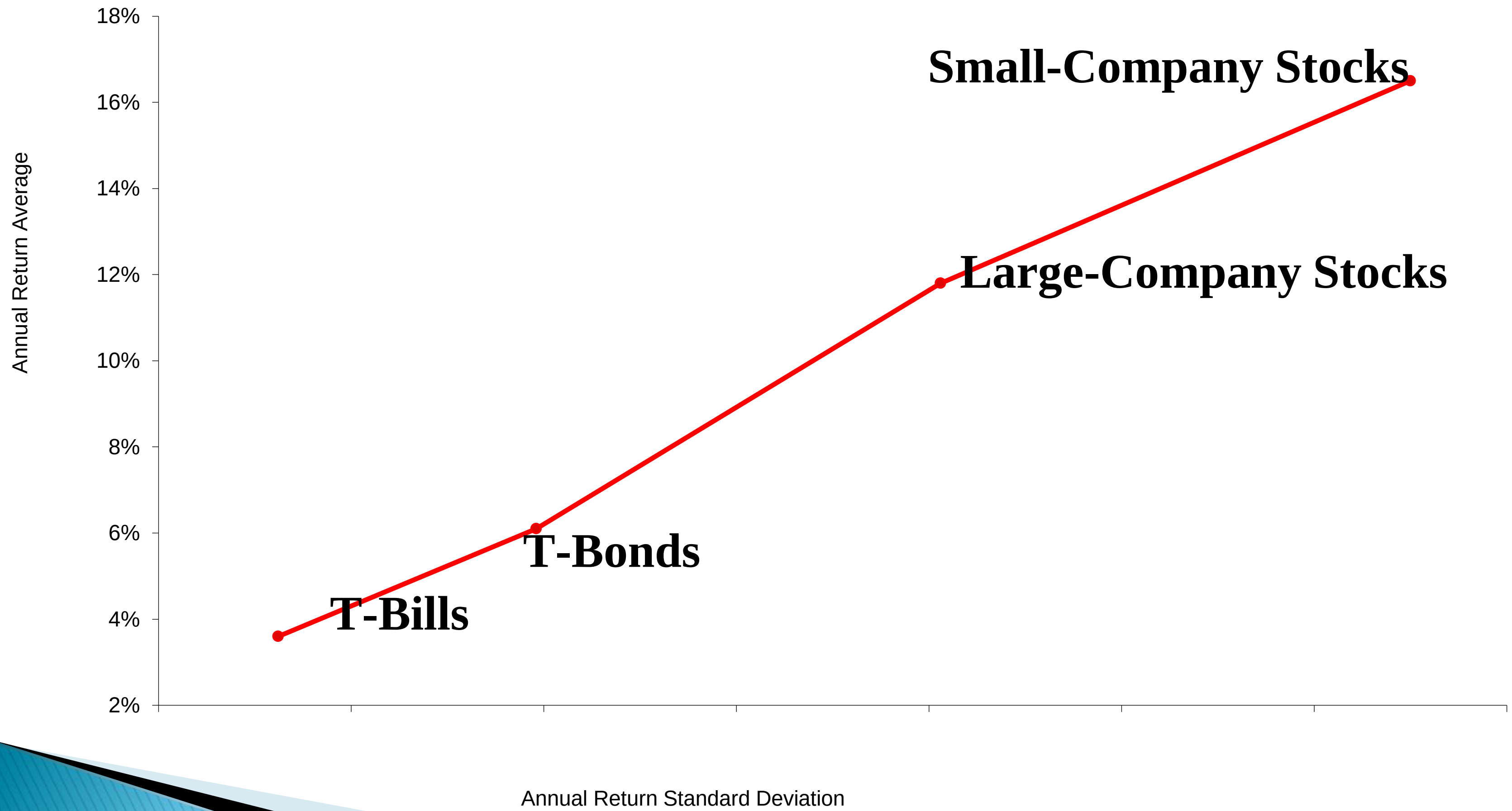
Dollar Return = Dividend + Change in Market Value

$$\text{percentage return} = \frac{\text{dollar return}}{\text{beginning market value}}$$

$$= \frac{\text{dividend} + \text{change in market value}}{\text{beginning market value}}$$

$$= \text{dividend yield} + \text{capital gains yield}$$

# The Risk-Return Tradeoff

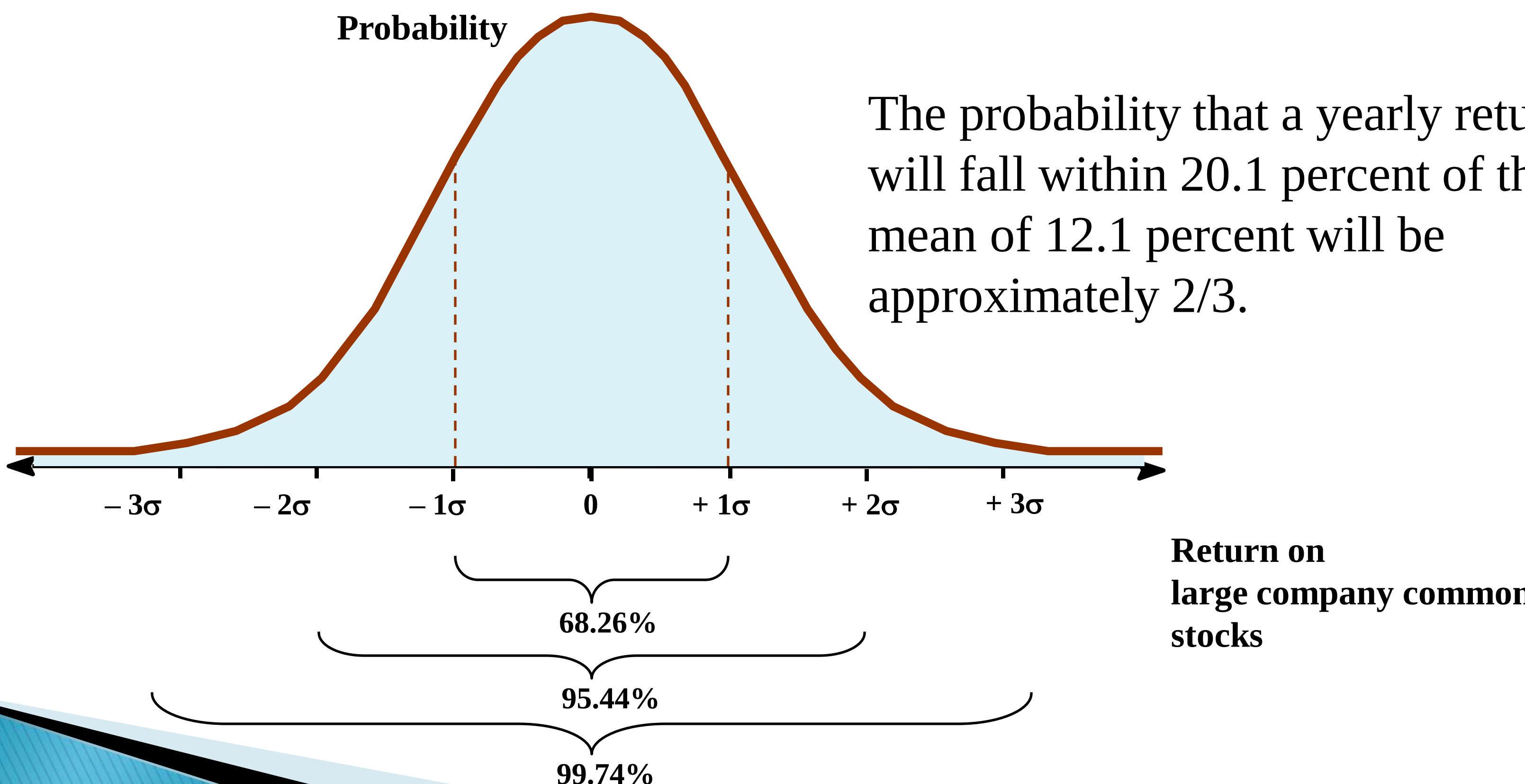


# Risk Statistics

- There is no universally agreed-upon definition of risk.
- The measures of risk that we discuss are variance and standard deviation.
  - The standard deviation is the standard statistical measure of the spread of a sample, and it will be the measure we use most of this time.
  - Its interpretation is facilitated by a discussion of the normal distribution.

# Normal Distribution

- A large enough sample drawn from a normal distribution looks like a bell-shaped curve.




The probability that a yearly return will fall within 20.1 percent of the mean of 12.1 percent will be approximately 2/3.

# Example – Return and Variance

| Year | Actual Return | Average Return | Deviation from the Mean | Squared Deviation |
|---|---|---|---|---|
| 1 | .15 | .105 | .045 | .002025 |
| 2 | .09 | .105 | -.015 | .000225 |
| 3 | .06 | .105 | -.045 | .002025 |
| 4 | .12 | .105 | .015 | .000225 |
| Totals | | | .00 | .0045 |

Variance = .0045 / (4-1) = .0015     Standard Deviation = .03873

# Expected Return, Variance, and Covariance

Consider the following two risky asset world. There is a 1/3 chance of each state of the economy, and the only assets are a stock fund and a bond fund.

| | | Rate of Return | |
|---|---|---|---|
| ***Scenario*** | ***Probability*** | ***Stock Fund*** | ***Bond Fund*** |
| Recession | 33.3% | -7% | 17% |
| Normal | 33.3% | 12% | 7% |
| Boom | 33.3% | 28% | -3% |

# Expected Return

| | Stock Fund | | Bond Fund | |
|---|---|---|---|---|
| **Scenario** | **Rate of Return** | **Squared Deviation** | **Rate of Return** | **Squared Deviation** |
| ***Recession*** | -7% | 0.0324 | 17% | 0.0100 |
| ***Normal*** | 12% | 0.0001 | 7% | 0.0000 |
| ***Boom*** | 28% | 0.0289 | -3% | 0.0100 |
| **Expected return** | 11.00% | | 7.00% | |
| **Variance** | 0.0205 | | 0.0067 | |
| **Standard Deviation** | 14.3% | | 8.2% | |

# Expected Return

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$$E(r_S) = \tfrac{1}{3} \times (-7\%) + \tfrac{1}{3} \times (12\%) + \tfrac{1}{3} \times (28\%)$$

$$E(r_S) = 11\%$$

# Variance

| Scenario | Stock Fund Rate of Return | Stock Fund Squared Deviation | Bond Fund Rate of Return | Bond Fund Squared Deviation |
|---|---|---|---|---|
| ***Recession*** | -7% | 0.0324 | 17% | 0.0100 |
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$$(-7\% - 11\%)^2 = .0324$$

# Variance

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$$.0205 = \frac{1}{3}(.0324 + .0001 + .0289)$$

# Standard Deviation

| | Stock Fund | | Bond Fund | |
|---|---|---|---|---|
| Scenario | Rate of Return | Squared Deviation | Rate of Return | Squared Deviation |
| ***Recession*** | -7% | 0.0324 | 17% | 0.0100 |
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$$14.3\% = \sqrt{0.0205}$$

# Covariance

| ***Scenario*** | ***Stock Deviation*** | ***Bond Deviation*** | ***Product*** | ***Weighted*** |
|---|---|---|---|---|
| ***Recession*** | -18% | 10% | -0.0180 | -0.0060 |
| ***Normal*** | 1% | 0% | 0.0000 | 0.0000 |
| ***Boom*** | 17% | -10% | -0.0170 | -0.0057 |
| **Sum** | | | | -0.0117 |
| **Covariance** | | | | -0.0117 |

"Deviation" compares return in each state to the expected return.

"Weighted" takes the product of the deviations multiplied by the probability of that state.

# Correlation

$$\rho = \frac{Cov(a,b)}{\sigma_a \sigma_b}$$

$$\rho = \frac{-.0117}{(.143)(.082)} = -0.998$$

# The Return and Risk for Portfolios

| | Stock Fund | | Bond Fund | |
|---|---|---|---|---|
| **Scenario** | **Rate of Return** | **Squared Deviation** | **Rate of Return** | **Squared Deviation** |
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Note that stocks have a higher expected return than bonds and higher risk. Let us turn now to the risk-return tradeoff of a portfolio that is 50% invested in bonds and 50% invested in stocks.

# Portfolios

| | Rate of Return | | | |
|---|---|---|---|---|
| ***Scenario*** | ***Stock fund*** | ***Bond fund*** | ***Portfolio*** | ***squared deviation*** |
| ***Recession*** | -7% | 17% | 5.0% | 0.0016 |
| ***Normal*** | 12% | 7% | 9.5% | 0.0000 |
| ***Boom*** | 28% | -3% | 12.5% | 0.0012 |
| | | | | |
| ***Expected return*** | 11.00% | 7.00% | 9.0% | |
| ***Variance*** | 0.0205 | 0.0067 | 0.0010 | |
| **Standard Deviation** | 14.31% | 8.16% | 3.08% | |

The rate of return on the portfolio is a weighted average of the returns on the stocks and bonds in the portfolio:

$$r_P = w_B r_B + w_S r_S$$

$$5\% = 50\% \times (-7\%) + 50\% \times (17\%)$$

# Portfolios

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The *expected* rate of return on the portfolio is a weighted average of the *expected* returns on the securities in the portfolio.

$$E(r_P) = w_B E(r_B) + w_S E(r_S)$$

$$9\% = 50\% \times (11\%) + 50\% \times (7\%)$$

# Portfolios

| | Rate of Return | | | |
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| ***Scenario*** | ***Stock fund*** | ***Bond fund*** | ***Portfolio*** | ***squared deviation*** |
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The variance of the rate of return on the two risky assets portfolio is

$$?_P^2 = (w_B ?_B)^2 + (w_S ?_S)^2 + 2(w_B ?_B)(w_S ?_S)?_{BS}$$

where $\rho_{BS}$ is the correlation coefficient between the returns on the stock and bond funds.

# Portfolios

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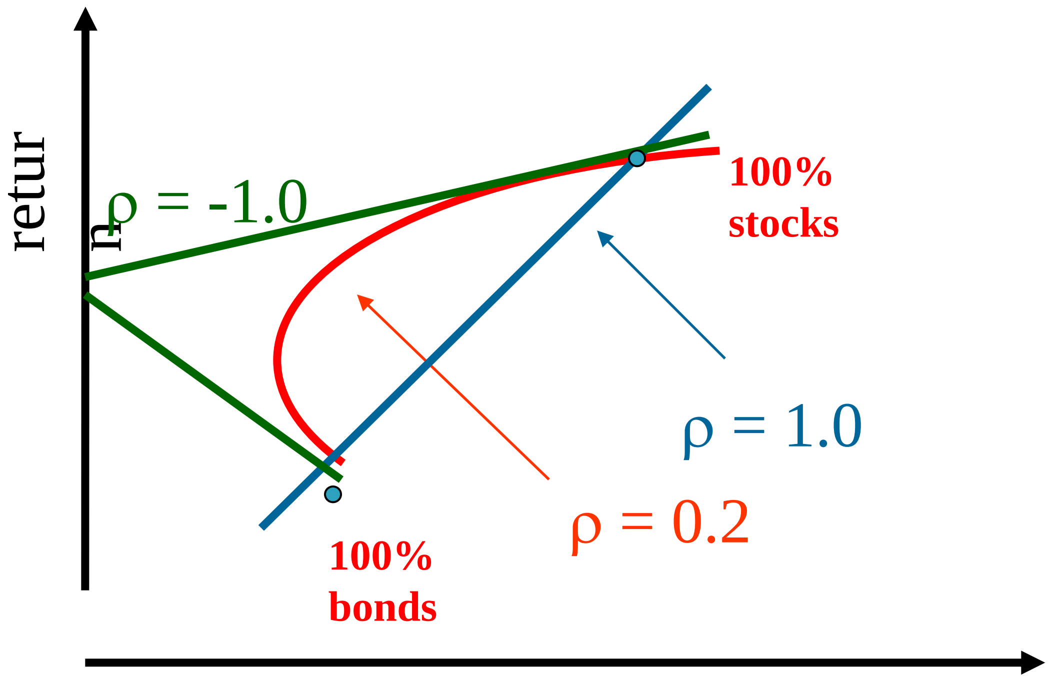
| ***Variance*** | 0.0205 | 0.0067 | 0.0010 | |
| **Standard Deviation** | 14.31% | 8.16% | 3.08% | |

Observe the decrease in risk that diversification offers.

An equally weighted portfolio (50% in stocks and 50% in bonds) has less risk than either stocks or bonds held in isolation. This is not always the case.

# Portfolios with Various Correlations

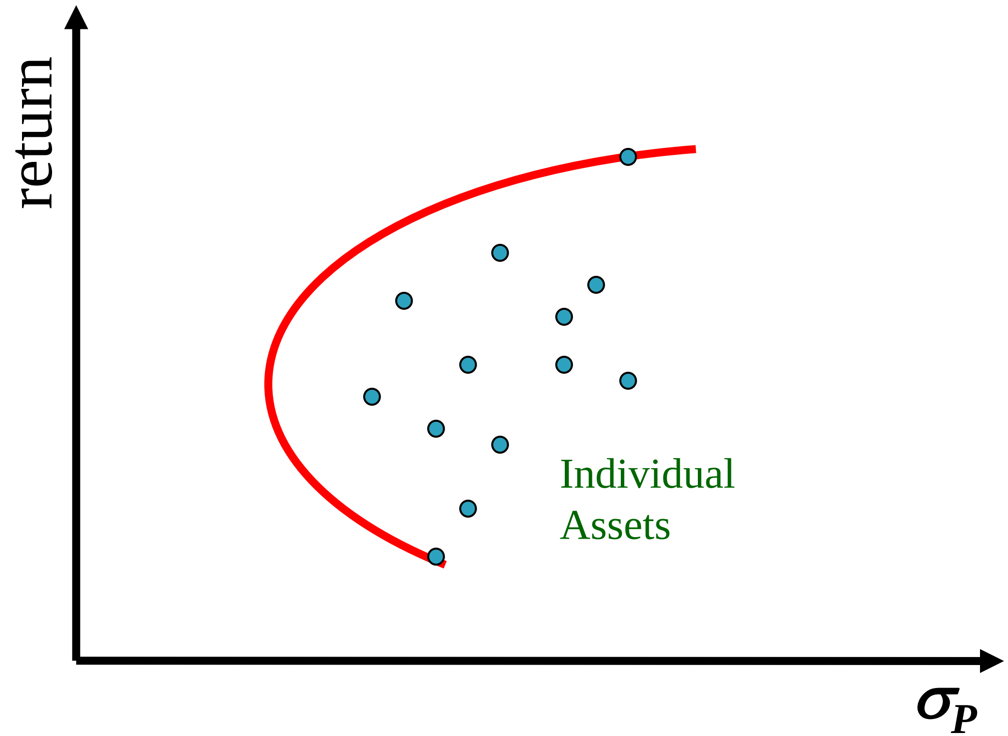



- Relationship depends on correlation coefficient

$$-1.0 \leq \rho \leq +1.0$$

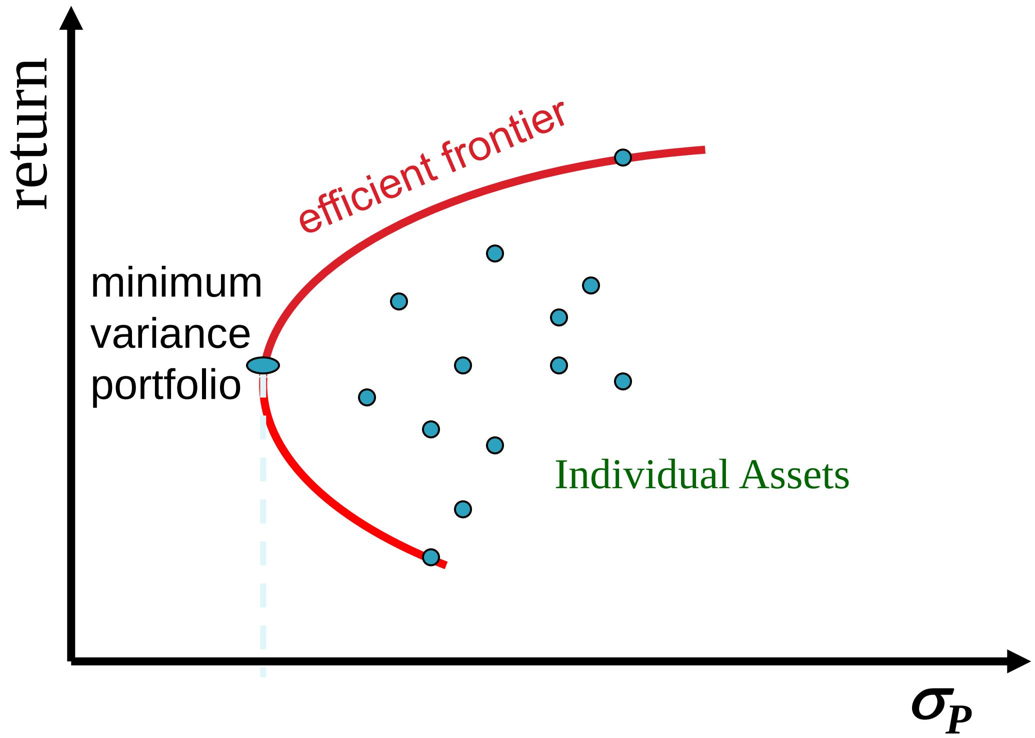
- If $\rho = +1.0$, no risk reduction is possible
- If $\rho = -1.0$, complete risk reduction is possible

# The Efficient Set for Many Securities




Consider a world with many risky assets; we can still identify the *opportunity set* of risk-return combinations of various portfolios.

# The Efficient Set for Many Securities



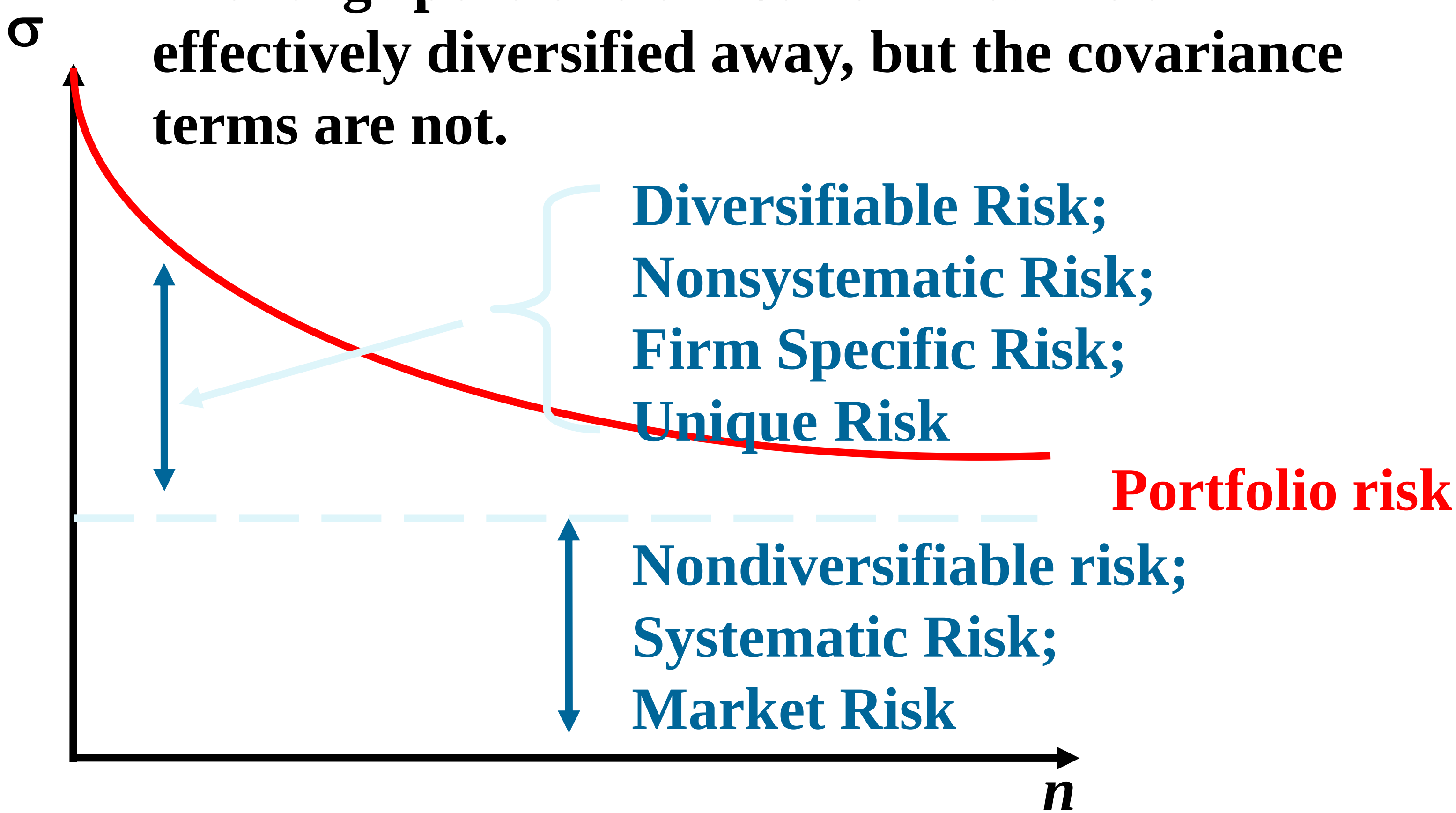

The section of the opportunity set above the minimum variance portfolio is the efficient frontier.

# Diversification and Portfolio Risk

- Diversification can substantially reduce the variability of returns without an equivalent reduction in expected returns.
- This reduction in risk arises because worse than expected returns from one asset are offset by better than expected returns from another.
- However, there is a minimum level of risk that cannot be diversified away, and that is the systematic portion.

# Portfolio Risk and Number of Stocks

**In a large portfolio the variance terms are effectively diversified away, but the covariance terms are not.**



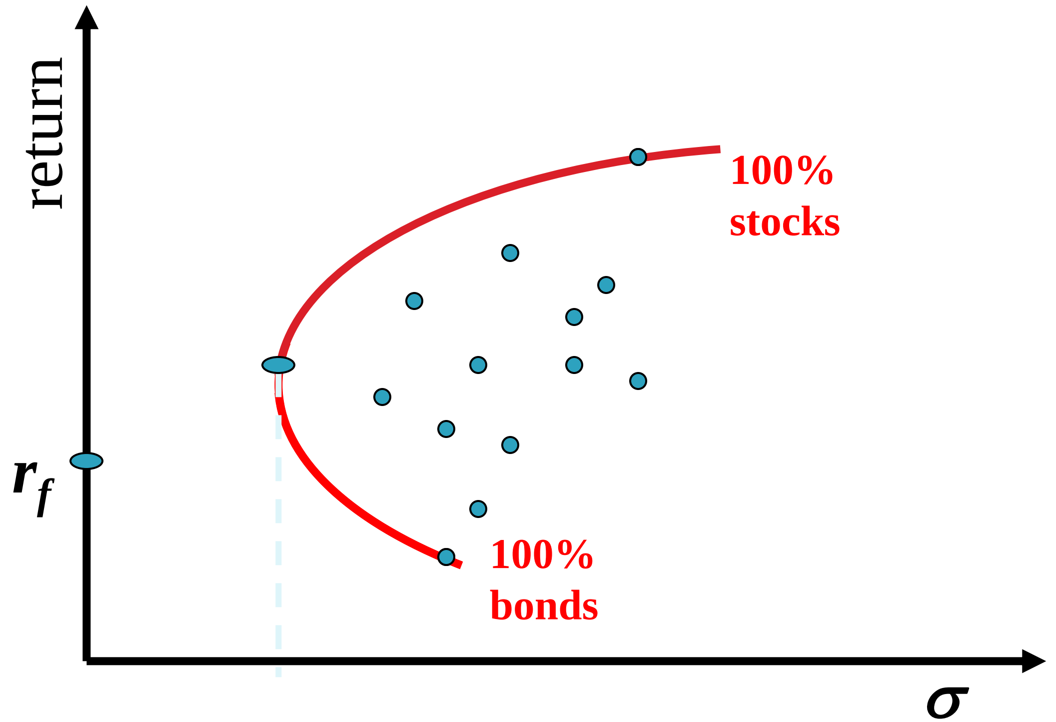
# Risk: Systematic and Unsystematic

- A *systematic risk* is any risk that affects a large number of assets, each to a greater or lesser degree.
- An *unsystematic risk* is a risk that specifically affects a single asset or small group of assets.
- Unsystematic risk can be diversified away.
- Examples of systematic risk include uncertainty about general economic conditions, such as GNP, interest rates or inflation.
- On the other hand, announcements specific to a single company are examples of unsystematic risk.

# Total Risk

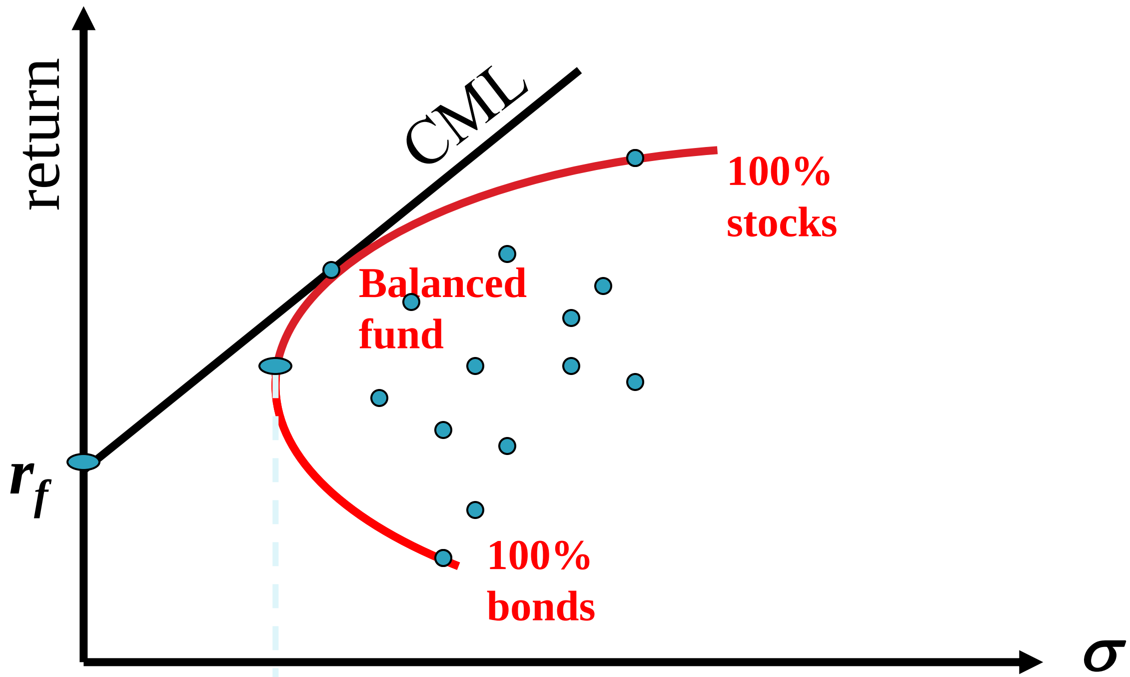
- Total risk = systematic risk + unsystematic risk
- The standard deviation of returns is a measure of total risk.
- For well-diversified portfolios, unsystematic risk is very small.
- Consequently, the total risk for a diversified portfolio is essentially equivalent to the systematic risk.

# Optimal Portfolio with a Risk-Free Asset



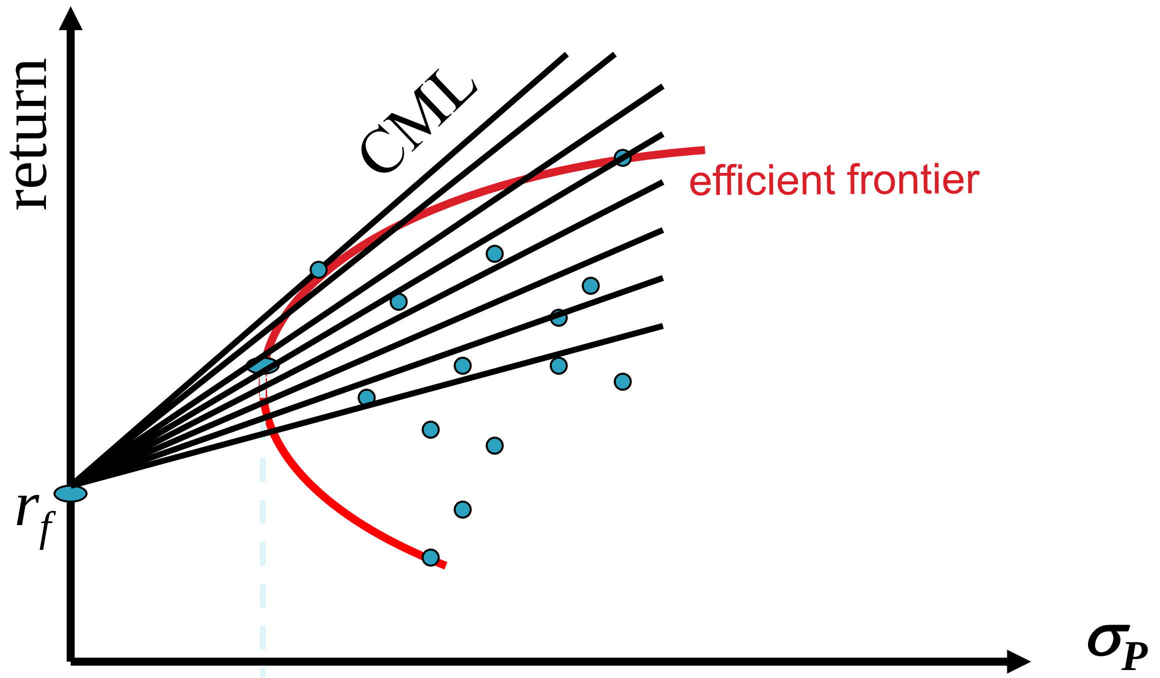

In addition to stocks and bonds, consider a world that also has risk-free securities like T-bills.

# Riskless Borrowing and Lending



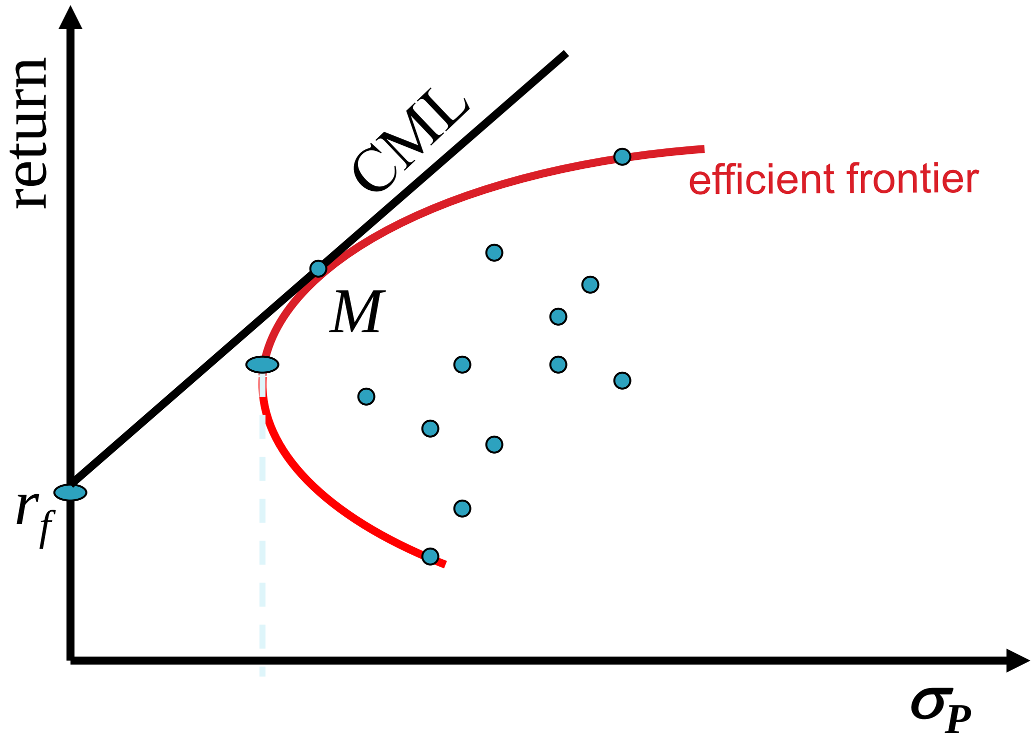

Now investors can allocate their money across the T-bills and a balanced mutual fund.

# Riskless Borrowing and Lending



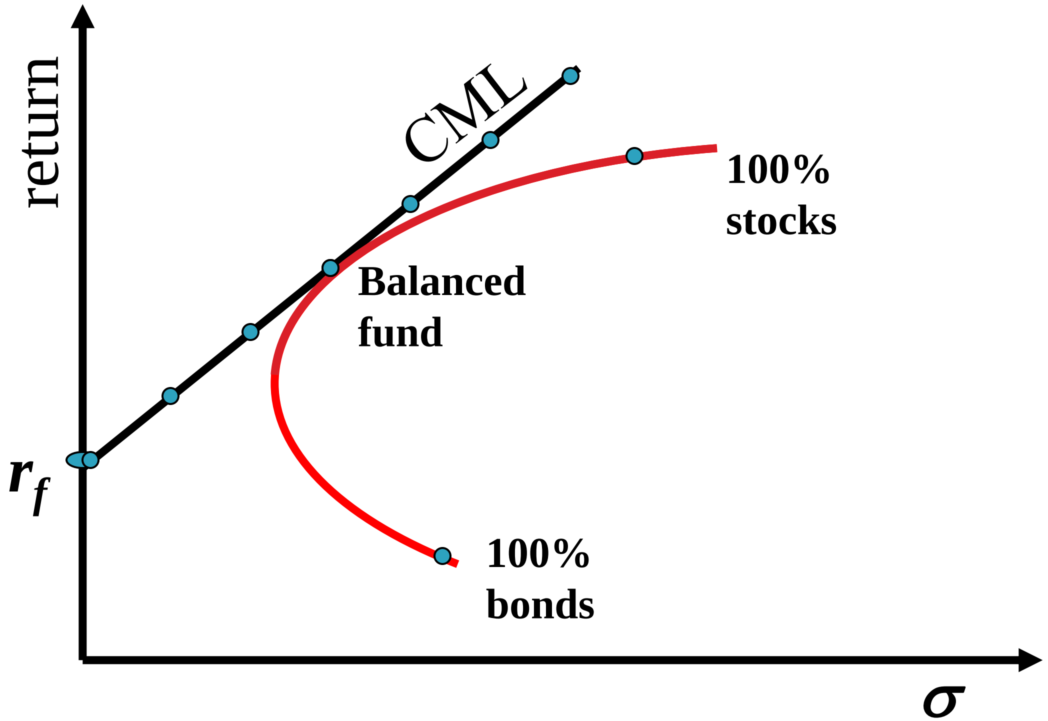

With a risk-free asset available and the efficient frontier identified, we choose the capital allocation line with the steepest slope.

# 11.8 Market Equilibrium




With the capital allocation line identified, all investors choose a point along the line—some combination of the risk-free asset and the market portfolio *M*. In a world with homogeneous expectations, *M* is the same for all investors.

# Market Equilibrium



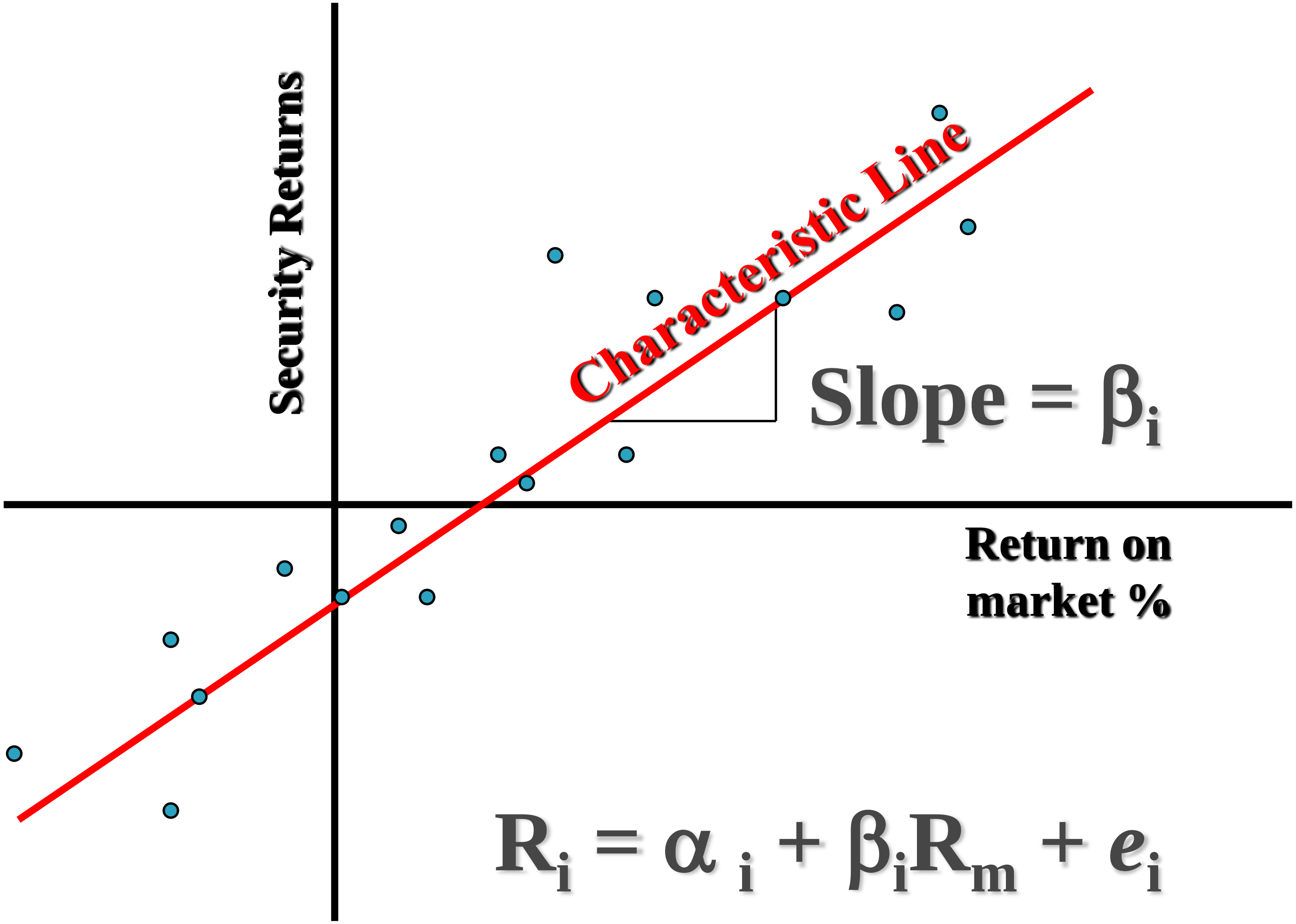

Where the investor chooses along the Capital Market Line depends on her risk tolerance. The big point is that all investors have the same CML.

# Risk When Holding the Market Portfolio

- Researchers have shown that the best measure of the risk of a security in a large portfolio is the *beta* (β) of the security.
- Beta measures the responsiveness of a security to movements in the market portfolio (i.e., systematic risk).

$$\beta_i = \frac{Cov(R_{i,} R_M)}{\sigma^2(R_M)}$$

# Estimating β with Regression




$$R_i = \alpha_i + \beta_i R_m + e_i$$

# The Formula for Beta

$$\beta_i = \frac{Cov(R_{i,} R_M)}{\sigma^2(R_M)} = \rho \frac{\sigma(R_i)}{\sigma(R_M)}$$

Clearly, your estimate of beta will depend upon your choice of a proxy for the market portfolio.

# Relationship between Risk and Expected Return (CAPM)

- Expected Return on the Market:

$$\bar{R}_M = R_F + \text{Market Risk Premium}$$

- Expected return on an individual security:

$$\bar{R}_i = R_F + \beta_i \times \underbrace{(\bar{R}_M - R_F)}_{\text{Market Risk Premium}}$$

*This applies to individual securities held within well-diversified portfolios.*

# Expected Return on a Security

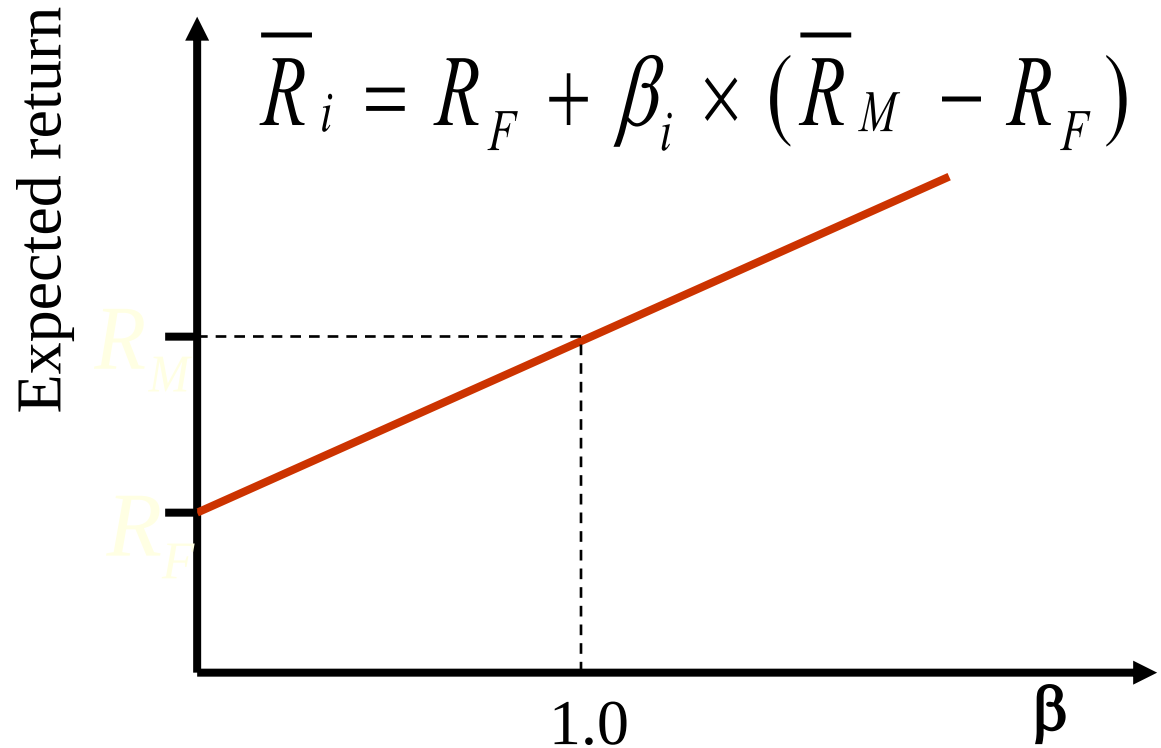
- This formula is called the Capital Asset Pricing Model (CAPM):

$$\bar{R}_i = R_F + \beta_i \times (\bar{R}_M - R_F)$$

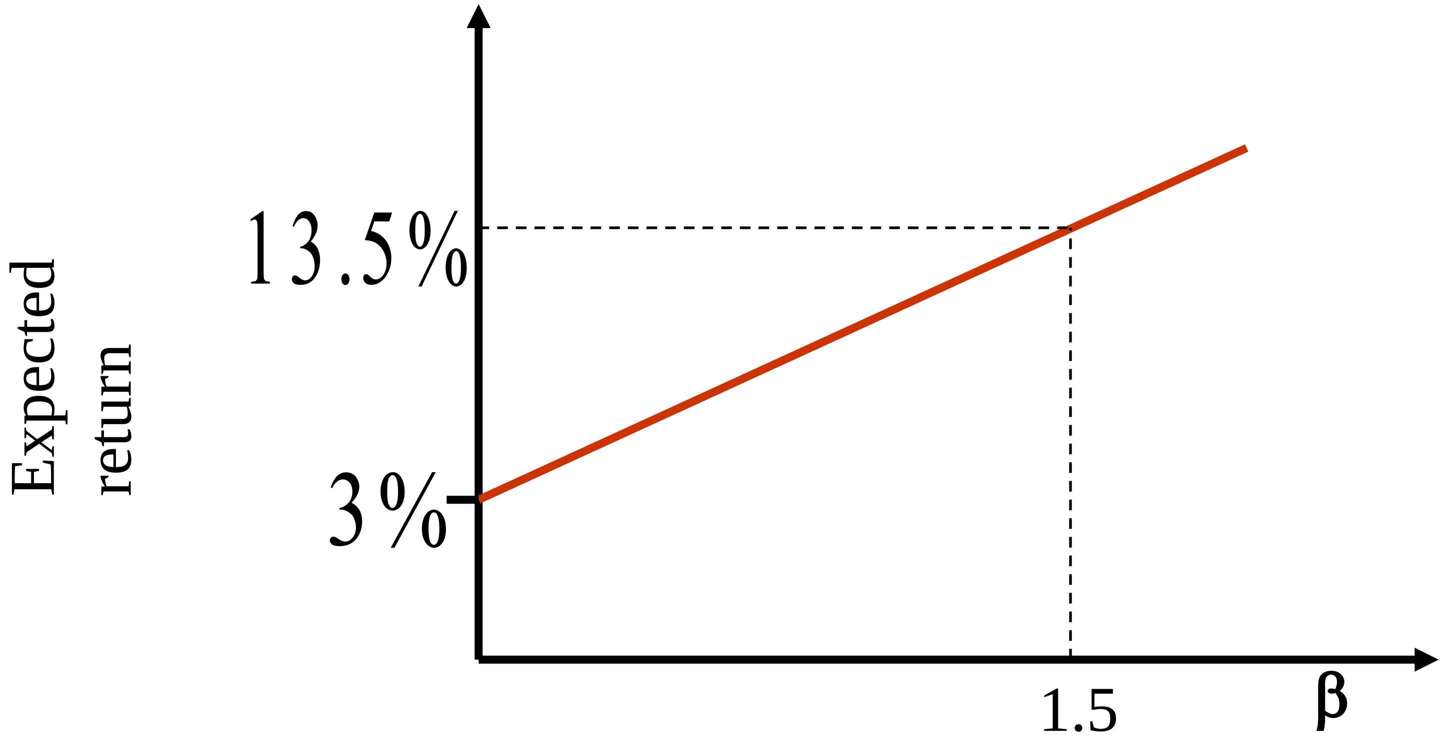
| | | | | | | |
|---|---|---|---|---|---|---|
| Expected return on a security | = | Risk-free rate | + | Beta of the security | × | Market risk premium |

- Assume $\beta_i = 0$, then the expected return is $R_F$.
- Assume $\beta_i = 1$, then $\bar{R}_i = \bar{R}_M$

# Relationship Between Risk & Return



# Relationship Between Risk & Return




$$\beta_i = 1.5 \quad R_F = 3\% \quad \bar{R}_M = 10\%$$

$$\bar{R}_i = 3\% + 1.5 \times (10\% - 3\%) = 13.5\%$$

Disclaimer: The content given in the slides have been collected from Corporate Finance by Ross, Westerfield, Jaffe, Jordan and Kakani (11Ed.) McGrawHill Education. This is pure prepared for the classroom discussions at BITS Pilani.